Benchmarking Causal Discovery Algorithms for Scale Free Graphs

Ritwick Banerjee[1]

[1] Institute of Health Informatics, University of Minnesota

Abstract :

Causal discovery algorithms have gained significant traction in various research domains, yet practitioners often lack intuitive understanding of underlying graph structures that substantially impact results. This study addresses this gap by empirically investigating different graph structures for multiple causal discovery approaches through comprehensive in-silico experimentation. We specifically focus on the Erdos Renyi (ER) and Scale Free (SF) graphs, which are two ends of the spectrum in terms of homogeneity and randomness, ER graphs are homogeneous and random whereas SF graphs are heterogeneous and structured. Using a systematic evaluation framework across varying network configurations (20 and 50 variables with average degrees of 2 and 6), we analyze F1 scores, precision, and recall metrics at different sample sizes to identify performance of causal methods on different graph types.

Our results demonstrate that given enough samples, constraint-based and score based methods converge to similar F1, Precision, and Recall scores for ER graphs, however, as the graph structure becomes more heterogeneous the metrics differ for each method. Among the methods considered, Best Order Score Search (BOSS) outperforms the other constraint based and score based algorithms. These findings provide evidence that BOSS is able to uncover causal relationships in Scale Free structures.The study contributes toward establishing transparent benchmark that enhance both the accessibility and reliability of causal discovery methods for researchers across disciplines.

Index Terms: Causal Discovery, Scale Free Graphs, BOSS

1 Study Goals and Motivation

As the field of Causal Discovery matures, its application has expanded from theoretical exercises to complex real-world domains such as neuroscience, bioinformatics, and scientometrics. While numerous algorithms exist to identify causal structures from observational data, their performance is often heavily dependent on the topological assumptions of the underlying data-generating process. A significant gap exists in understanding how these methods perform when the underlying structure deviates from random connectivity to more complex, organized hierarchies.

Motivation: The primary motivation for this simulation study stems from the observation that real-world networks rarely resemble random graphs. In nature and society, connectivity is frequently governed by preferential attachment, leading to Scale-Free networks characterized by power-law degree distributions.

- Brain Networks: In neuroscience, functional and structural connectivity often involves "hub" nodes (highly connected regions) and "sink" nodes, facilitating efficient information transfer.
- Citation Networks: In academia, citation graphs naturally evolve into scale-free structures where a minority of papers receive the majority of citations.

If a Causal Discovery method is biased towards, or implicitly assumes, a random distribution of edges such as in Erdős-Rényi graphs, its application to these specialized domains may yield spurious edges or miss critical hub connections. Therefore, relying on methods validated primarily on random graphs poses a significant risk for researchers attempting to map the causal architecture of the brain or information flow in literature.

Study Goals: The goal of this report is to identify the most robust causal discovery method for networks exhibiting specific topological characteristics. We conduct a comprehensive in-silico experiment to benchmark various algorithms against four distinct ground-truth structures:

- Erdős-Rényi Graphs (as a baseline control).
- Scale-Free In-Degree (simulating sink-heavy structures).
- Scale-Free Out-Degree (simulating hub-heavy structures).
- Scale-Free Both (simulating complex, high-density hub-and-sink structures).

By systematically varying the generative structure, we aim to isolate the effect of network topology on algorithm performance. Ultimately, this study seeks to provide prescriptive guidance for practitioners in neuroscience and network science, ensuring that the chosen causal discovery method is topologically appropriate for the domain in question.

1.1 Literature Review

The evaluation of causal discovery algorithms heavily relies on synthetic data simulations, where the ground truth is known, and performance metrics can be rigorously calculated. Historically, the vast majority of these in-silico experiments have defaulted to generating ground truth structures using random graph models, most notably the Erdős-Rényi (ER) model. While mathematically convenient, ER graphs fail to capture the heterogeneous connectivity patterns, such as hubs and sinks, that are ubiquitous in real-world systems like biological networks (Barabási et al., 1999) and citation graphs.

Despite the known topological discrepancy between standard benchmarks and real-world applications, there remains a paucity of research explicitly investigating how the structural properties of the data-generating mechanism influence algorithm performance. To our knowledge, only one recent study has directly addressed the generation and validation of scale-free Directed Acyclic Graphs (DAGs) for this purpose. (Andrews et al., 2024) introduced a novel DAG adaptation of the "onion method," a framework capable of enforcing scale-free properties in both in-degree and out-degree distributions while maintaining acyclicity. In their preliminary simulations, (Andrews et al., 2024) demonstrated that scale-free graphs present a significantly more challenging optimization landscape for causal discovery than their random counterparts, suggesting that current state-of-the-art results may be optimistic when applied to non-random domains.

This paper seeks to bridge this gap by conducting a comprehensive topological benchmark. Building directly upon the framework proposed by (Andrews et al., 2024), we systematically evaluate causal discovery performance across four distinct structural regimes: Erdős-Rényi (baseline), Scale-Free In-Degree (sink-dominated), Scale-Free Out-Degree (hub-dominated), and Scale-Free Both. By extending the analysis to include a diverse suite of algorithms—including permutation-based (BOSS), constraint-based (PC), greedy score-based (FGES), and continuous optimization (DAGMA) methods—we aim to provide robust, prescriptive guidance for practitioners operating in topologically complex domains.

2 Background

2.1 Directed Acyclic Graphs

A *directed acyclic graph* (DAG) is an edge vertex graph whose vertices and edges represent variables and conditional independence statements. We use the notation $\mathcal{G} = (V, E)$ where $\mathcal{G}$ is a graph, V is a vertex set, and E is an edge set. In a DAG, the edge set is comprised of ordered pairs that represent directed edges[1] and contains no directed cycles.

Let $\mathcal{G} = (V, E)$ be a DAG and $v \in V$:

$$\mathrm{pa}_{\mathcal{G}}(v) \equiv \{w \in V \ : \ v \leftarrow w \text{ in } \mathcal{G}\}$$
$$\mathrm{ch}_{\mathcal{G}}(v) \equiv \{w \in V \ : \ v \rightarrow w \text{ in } \mathcal{G}\}$$

are the *parents* and *children* of v.

A DAG encodes graphical conditional independence between disjoint subsets $A, B, C \subseteq V$ which can be read off of the DAG using *d-separation* and is denoted $A \perp\!\!\!\perp B \mid C \ [\,\mathcal{G}\,]$ (Dawid, 1979; Pearl, 1988; Lauritzen, 1996). This criterion admits the concept of a *Markov equivalence class* (MEC) which—in the context of this paper—is a collection of DAGs that represent the same conditional independence relations (Spirtes et al., 1993).

2.2 Causal Discovery

DAGs are connected to causality by the *causal Markov* and *causal faithfulness* assumptions. A DAG is *causal* if it describes the true underlying causal process by placing a directed edge between a pair of variables if and only if the former directly causes the latter (Spirtes et al., 1993).

Causal Markov: If $\mathcal{G} = (V, E)$ is causal for a collection of random variables X indexed by V with probability distribution P, then for disjoint subsets $A, B, C \subseteq V$:

$$A \perp\!\!\!\perp B \mid C \ [\,\mathcal{G}\,] \quad \Rightarrow \quad A \perp\!\!\!\perp B \mid C \ [\,P\,].$$

More generally, this implication is called the *Markov property* and we say DAGs satisfying this property *contains* the distribution.

Causal faithfulness: If $\mathcal{G} = (V, E)$ is causal for a collection of random variables X indexed by V with probability distribution P, then for disjoint subsets $A, B, C \subseteq V$:

$$A \perp\!\!\!\perp B \mid C \ [\,P\,] \quad \Rightarrow \quad A \perp\!\!\!\perp B \mid C \ [\,\mathcal{G}\,].$$

Under the causal Markov and causal faithfulness assumptions, finding the MEC of the causal DAG is equivalent to identifying the simplest DAG(s) in terms of edge count that contains the data generating distribution. Causal discovery algorithms automate this search process.

2.3 The Bayesian Information Criterion and Fisher Z Test

Causal Discovery algorithms are generally characterized as score-based or constraint-based. Both approaches assess conditional independence relationships in that data—score-based methods with a goodness-of-fit score and constraint-based methods with conditional independence tests. In this paper we focus on the Bayesian Information Criterion (BIC) (Schwarz, 1978; Haughton, 1988) for score-based evaluation and the Fisher Z Test (Spirtes et al., 1993) for conditional independence testing in Gaussian models.

For a Gaussian model characterized with DAG $\mathcal{G}$ the BIC on X is defined as:

$$\begin{aligned}\mathrm{BIC}(\mathcal{G}, X) &= \ell(\hat{\theta}_{\mathcal{G}}) - \frac{|E|}{2}\log(n) \\ &= \sum_{v \in V} \ell(\hat{\theta}_{v|\mathrm{pa}_{\mathcal{G}}(v)}) - \frac{|\mathrm{pa}_{\mathcal{G}}(v)|}{2}\ \log(n)\end{aligned}$$

where $\hat{\theta}_*$ is the MLE.

When employing constraint-based approaches, we rely on statistical tests to determine if two variables X and Y are conditionally independent given a conditioning set Z, denoted as $X \perp\!\!\!\perp Y \mid Z$. For Gaussian distributed data, this is equivalent to testing if the partial correlation coefficient $\rho_{XY \cdot Z}$ is zero. The sample partial correlation $\hat{\rho}_{XY \cdot Z}$ is computed from the inverse covariance matrix (precision matrix). To test the null hypothesis $H_0 : \rho_{XY \cdot Z} = 0$, we apply the Fisher Z-transformation:

$$z(\hat{\rho}_{XY \cdot Z}) = \frac{1}{2}\ln\left(\frac{1 + \hat{\rho}_{XY \cdot Z}}{1 - \hat{\rho}_{XY \cdot Z}}\right).$$

Under the null hypothesis, the statistic $\sqrt{n - |Z| - 3} \cdot z(\hat{\rho}_{XY \cdot Z})$ asymptotically follows a standard normal distribution. We reject the null hypothesis of independence (and infer an edge or dependency exists) at significance level α if:

$$\sqrt{n - |Z| - 3}\ \ |z(\hat{\rho}_{XY \cdot Z})| > \Phi^{-1}\left(1 - \frac{\alpha}{2}\right)$$

where n is the sample size, $|Z|$ is the size of the conditioning set, and Φ^{-1} is the inverse cumulative distribution function of the standard normal distribution.

2.4 Metrics

We use the following metrics to evaluate the learnt graphs by using an ensemble method to compare against the true data generating graph. The ensemble estimate is calculated by thresholding predicted probabilities at 0.5, i.e., considering no edge between two node pairs if the predicted probabilities are lesser than 0.5. For metrics that hint at calibration and reliability we calculate them via calculating an edge frequency table which stores the probabilites of two nodes being adjacent to each other. Then we compute the following metrics (Brier and ECE) using the edge frequency table.

2.4.1 Brier Scores In order to evaluate the calibration of the resampling techniques, we treat the resampling adjacency frequencies as predictive probability, and evaluate their predictive ability using

[1] The edge $v \leftarrow w$ corresponds to the order pair (v, w).

the Brier score (Brier, 1950), a standard tool for evaluating the performance for binary events. The Brier score is simply the mean squared error between the predicted probability and the observed outcomes:

$$\text{Brier Score} = \frac{1}{n}\sum_{s=1}^{n}(f_s - o_s)^2 \tag{1}$$

where n is the sample size, fs is the forecast probability of the event in sample s, and os is the observed outcome in sample s, with 0 meaning the event did not occur and 1 meaning the event did occur. Since this is a measure of error, larger values are worse: the best Brier score possible is 0, while the worst score possible is 1.

2.4.2 Expected Calibration Error While Brier scores are commonly used to evaluate the calibration of probabilistic models, they provide an incomplete assessment of model calibration on their own. To gain a more comprehensive understanding, Expected Calibration Error (ECE) (Pakdaman Naeini et al., 2015) is often used in conjunction with Brier scores. ECE measures the discrepancy between predicted probabilities and observed outcomes by dividing the predictions into $\boldsymbol{K}$ equally spaced bins based on their confidence scores. Mathematically, ECE is defined as:

$$\text{ECE} = \sum_{i=1}^{K} P(i) \cdot |o_i - e_i| \tag{2}$$

where o_i is the true fraction of positive instances in bin i, e_i is the mean of the post-calibrated probabilities for the instances in bin i, and P(i) is the empirical probability (fraction) of all instances that fall into bin i. Lower ECE values indicate better calibration, as they suggest that the predicted probabilities more accurately reflect the observed frequencies of the positive class.

2.4.3 Precision, Recall, and F1-score We report Precision, Recall, and F1-score to capture model performance (Sokolova et al., 2009). These metrics offer a more detailed understanding of how well a model identifies true positive instances while minimizing false positives and false negatives.

Precision measures the proportion of correctly predicted positive instances out of all instances predicted as positive. It is defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

where TP denotes true positives and FP denotes false positives. High precision indicates a low rate of false positives, ensuring that the positive predictions made by the model are reliable.

Recall, measures the proportion of actual positive instances that were correctly identified by the model. It is defined as:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4}$$

where FN denotes false negatives. High recall ensures that most of the actual positive instances are captured by the model, minimizing missed detections.

To balance the trade-off between Precision and Recall, the F1-score is often used. The F1-score is the harmonic mean of Precision and Recall, providing a single composite metric that captures both aspects of model performance:

$$\text{F1 score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

The F1-score ranges from 0 to 1, with higher values indicating better overall model performance. It is particularly useful in scenarios with class imbalance, where optimizing for either Precision or Recall alone may lead to suboptimal model behavior.
sectionData Description

2.5 Overview

To simulate the data we used the DaO method (Andrews et al., 2024) to simulate data involving four graph types, Erdos Renyi and Scale Free in, Scale Free out, and Scale Free both Graphs.

The data were generated from a causal graphical model, followed by repeated sub-sampling to create large collections of datasets. Each resampled dataset was analyzed using a causal discovery algorithm namely Best Order Score Search (BOSS), Fast Greedy Equivalence Search (FGES), Peter Clark (PC), DAGMA with a default parameters, resulting in a corresponding large collection of graphs.

For each pair of variables, average degree, sample size, and algorithm we calculated the proportion of times a specific relationship was represented by an edge (or lack of one) between them across the generated graphs. These proportions were then evaluated as predictive probabilities of whether an edge existed in the underlying data-generating model. We assessed the predictive probabilities (selecting a threshold of 0.5 for Precision, Recall, and F1 score) using several metrics, including Precision, Recall and F1 score, comparing them to the true graphs that generated the data.

The entire process can be visualized by the picture below:

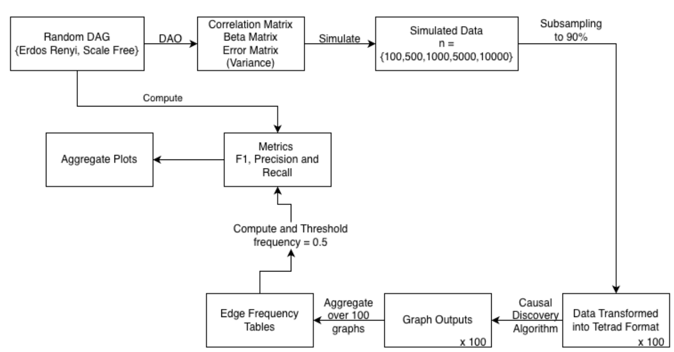

Figure 1. Simulation Flow Diagram

2.6 Data

To evaluate the performance of various algorithms, we performed an in-silico experiment involving simulations.

Random DAG: We generated random Erdős Rényi and scale free DAGs with $variables \in \{20, 50\}$ and $average\ degree \in \{2, 6\}$. For stability purposes, we generate 250 true graphs, for $variable = 20$, and 50 true graphs for $variable = 50$.

Correlation Matrix: The DAG is then used to compute a correlations matrix (and therefore a beta matrix and covariance matrix for additive error terms) using the DaO method (Andrews et al., 2024).

Simulated Data: Datasets are simulated using the beta matrix and covariance matrix for additive error terms. We generated datasets for several sample sizes such that $sample\ sizes = \{100, 500, 1000, 5000, 10000\}$.

3 Methods

The causal discovery algorithms were applied to each combination of the simulation parameters as mentioned under the 2.6 using Jpype, the Python wrapper of Tetrad (Joseph D Ramsey et al., 2018; Joseph D. Ramsey et al., 2023). One of the algorithm used in this paper is the Best Order Score Search (BOSS) (Andrews et al., 2023). BOSS greedily searches over permutations of variables, using GSTs (Grow-Shrink Trees) to construct and score DAGs from permutations. GSTs efficiently cache scores to eliminate redundant calculations. BOSS achieves state-of-the-art performance in accuracy and execution time, comparing favorably to a variety of combinatorial and gradient-based learning algorithms under a broad range of conditions.

The PC algorithm (Spirtes et al., 1993) is a foundational constraint-based method used to infer causal structure. It begins with a complete undirected graph and iteratively removes edges between variables that are statistically independent conditional on a subset of neighbors. The search is computationally efficient as it starts with low-order conditioning sets and incrementally increases their size. Once the graph skeleton is established, PC identifies v-structures (colliders) and applies a set of orientation propagation rules (Meek rules) to return a CPDAG representing the equivalence class of the underlying causal structure.

We employ Fast Greedy Equivalence Search (FGES) (J. Ramsey et al., 2017), an optimized score-based algorithm that acts as a scalable implementation of the Greedy Equivalence Search (GES). FGES operates in the space of Markov equivalence classes, represented by Completed Partially Directed Acyclic Graphs (CPDAGs), rather than searching over individual DAGs. It utilizes a two-phase greedy strategy: a forward phase that adds edges to maximize the score, followed by a backward phase that prunes edges to further optimize the model. By leveraging the local decomposability of scores (such as BIC) and parallelization, FGES achieves high computational efficiency on large datasets while maintaining asymptotic correctness.

This study also includes DAGMA (Bello et al., 2022), a gradient-based approach that formulates structure learning as a continuous optimization problem. Unlike combinatorial search methods, DAGMA utilizes a novel characterization of acyclicity involving the log-determinant of the inverse Hessian matrix, which allows the discrete constraint of acyclicity to be relaxed into a smooth penalty term. This formulation enables the use of standard stochastic gradient descent to optimize the weighted adjacency matrix. DAGMA is noted for its ability to handle non-linear relationships and its rapid convergence compared to prior continuous optimization techniques like NOTEARS.

Causal discovery algorithms are often difficult to trust in real-world applications due to concerns about the reliability of their outputs. A commonly accepted strategy to increase confidence in the results is to evaluate the stability of the learned causal graphs under resampling. To assess the robustness of our findings, We use a resampling-based approach that applies a 90% subsample. Since the primary objective of this project is to benchmark causal discovery methods on different graph types we run the algorithms on our simulated data with default settings.

After the search completed, the graphs learnt for each iteration of the 100 resamples, were converted into edge frequency table averaging over the graphs on the basis of how frequently an edge existed in those 100 resampled graphs for each original file (50 for 50 variables and 250 for 20 variables). These frequency tables were then compared with the true graphs to compute metrics such as F1 Score, Precision, and Recall with a threshold of 0.5.

3.1 Settings

In this study, a comprehensive grid search was conducted over a series of graph types to determine the best algorithm that is able to uncover the true causal graph. The penalty discount, a tuning parameter for the score based searches were kept at the default value of 2. The alpha value for the independence test which is intuitively the same as the penalty discount in terms of enforcing sparsity in the learnt graph was kept at a value of 0.01.

To assess the stability of the causal discovery outputs, the bootstrap method with adjusted sample sizes was implemented as the resampling technique. This approach allowed for repeated sampling with replacement, thereby providing insight into the variability of edge inclusion under different penalty regimes.

All algorithms were supplied with a precomputed correlation matrix, which is a standard input for the listed causal discovery methods. The correlation structure was estimated using the DaO method, ensuring that the matrix accurately reflected the underlying variable inter dependencies.

Model selection was driven by the structural equation model Bayesian Information Criterion (SEM-BIC) score, which was optimized at each iteration of the search algorithm for the score based methods (FGES and BOSS). The SEM-BIC criterion balanced model fit and complexity, facilitating the identification of parsimonious causal structures. For the constraint-based approach, we used the Fisher's Z test to test for conditional independencies

The number of random restarts ("starts") is not formally treated as a tuning parameter within this work, it was maintained at the software default of one to preserve computational consistency across penalty settings.

3.2 Choices

In configuring the simulated data environments and algorithmic parameters for this study, the following decisions were made to ensure a comprehensive evaluation of causal discovery performance across diverse scenarios:

Graph Type, Variable Count, Average Degree, and Sample Size Four canonical network structures were selected to emulate both purely random and empirically motivated dependency patterns. The Erdős–Rényi (ER) model, characterized by n nodes and M uniformly random edges, served to benchmark performance on archetypal random graphs. In contrast, Scale-Free (SF) networks, which exhibit heavy-tailed degree distributions akin to many real-world systems, were included to assess algorithmic robustness under more heterogeneous connectivity. Variable counts were systematically varied to encompass both "thin" datasets (fewer variables) and "fat" datasets (greater dimensionality), while average degrees spanned from sparse to dense regimes. Sample sizes were likewise chosen to span small, moderate, and large cohorts, reflecting the range of dataset lengths commonly explored in causal discovery literature.

Algorithm Selection To ensure a comprehensive evaluation of causal discovery performance across scale-free topologies, we selected four algorithms—BOSS, FGES, PC, and DAGMA. These methods were chosen to represent distinct theoretical paradigms within the field, ranging from classical constraint-based inference to modern continuous optimization. This diversity allows us to isolate how different search strategies cope with the specific challenges of hub nodes and power-law distributions.

- **BOSS (Permutation-based Score Search):** We selected BOSS (Andrews et al., 2023) as the representative for state-of-the-art permutation-based methods. Recent empirical studies suggest that searching over the space of variable orderings, rather than DAGs directly, yields superior edge recovery rates. Given that scale-free networks contain high-density hubs which are often difficult to orient correctly, BOSS's demonstrated ability to maximize edge recall makes it a critical candidate for benchmarking against complex ground truths.
- **FGES (Greedy Score-based Search):** As an optimized implementation of the Greedy Equivalence Search, FGES (J. Ramsey et al., 2017) serves as the primary baseline for score-based learning. It is widely regarded as the industry standard for scalability and reliability. Including FGES allows us to benchmark whether modern approaches offer statistically significant improvements over established greedy search heuristics when applied to non-random, scale-free topologies.
- **PC (Constraint-based Search):** We include the PC algorithm (Spirtes et al., 1993) as the foundational baseline for constraint-based methods. Its performance is particularly relevant to this study because its reliance on conditional independence testing (Fisher Z-test) faces unique challenges in scale-free networks. Specifically, the presence of high-degree "hub" nodes requires conditioning sets of increasing size, which can reduce statistical power and introduce errors. Evaluating PC allows us to quantify the degradation of constraint-based inference in the presence of hubs and sinks.
- **DAGMA (Continuous Optimization):** Finally, we selected DAGMA (Bello et al., 2022) to represent the emerging paradigm of differentiable causal discovery. Unlike the combinatorial nature of BOSS, FGES, and PC, DAGMA treats structure learning as a continuous optimization problem solvable via gradient descent. This inclusion is motivated by the need to understand if the relaxation of the acyclicity constraint—a technique that has shown promise in general settings—remains robust when the underlying data-generating process is strictly governed by scale-free mechanisms.

Penalty Discount and Number of Starts Since the primary goal of this work is to evaluate edge recovery in different graph types, the penalty discount or the alpha level of the conditional independence tests were kept at default values of 2 and 0.01 respectively. The number of random restarts ("starts") was held at the default value of one, consistent with standard practice in instructional settings and to isolate the impact of the penalty parameter without introducing additional sources of variance.

Scoring Criterion The SEM-BIC score was selected as the optimization target, reflecting both the linear Gaussian assumptions underlying the simulated datasets and the compatibility of the SEM-BIC criterion within the Tetrad framework. SEM-BIC's balance of model fit and complexity, combined with its prevalence in pedagogical examples, rendered it the natural choice for this investigation.

Implementation Details A high-performance C implementation of the BOSS algorithm, developed by Dr. Bryan Andrews, was utilized in place of the standard Java-based Tetrad version. This implementation, interfaced along JPype to access certain Tetrad features, afforded substantial gains in computational speed without compromising algorithmic fidelity. The other algorithms were implemented in accordance with the Tetrad framework except DAGMA, which had its own package in python.

Performance Metrics To quantify the accuracy of the recovered graph structures, we computed the F1 score alongside its constituent metrics of precision and recall. These metrics collectively capture the trade-offs between correctly identified edges and erroneous inclusions or omissions.

Adjacency Versus Edge-Frequency Analysis While adjacency matrices provide a binary indicator of inferred connections, more nuanced insights can be gleaned from edge-frequency tables generated across bootstrap iterations. Although implementation of frequency-based summaries proved technically challenging until recently, both binary adjacency and edge-frequency approaches are ultimately reported to offer a richer understanding of edge stability under resampling.

4 Results

Here are the Results:

Reporting a subset of results here, The exhaustive set of plots are not presented because of redundancy of information, however, they are available on request.

We report the performance of BOSS, FGES, PC, and DAGMA across four graph topologies: Erdős-Rényi (ER), Scale-Free Out-Degree (SF-Out), Scale-Free In-Degree (SF-In), and Scale-Free Both (SF-Both). All experiments were conducted on graphs with 50 variables and an average degree of 6, across sample sizes ranging from 10^2 to 10^4.

4.1 Performance on Erdős-Rényi (ER) Graphs

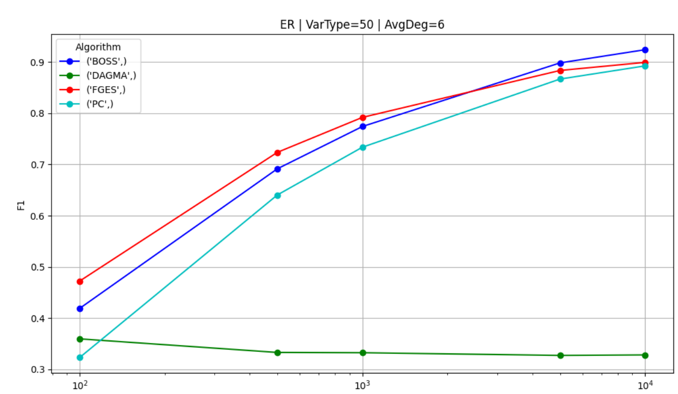

Figure 2. Erdos Renyi Graph, 50 Variables, Average Degree 6. F1 plot

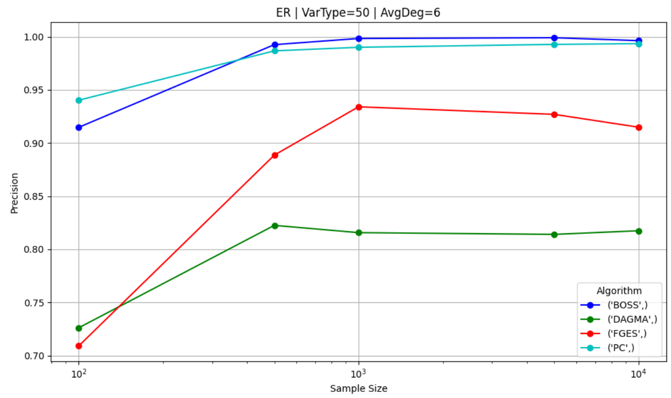

Figure 3. Erdos Renyi Graph, 50 Variables, Average Degree 6. Precision plot

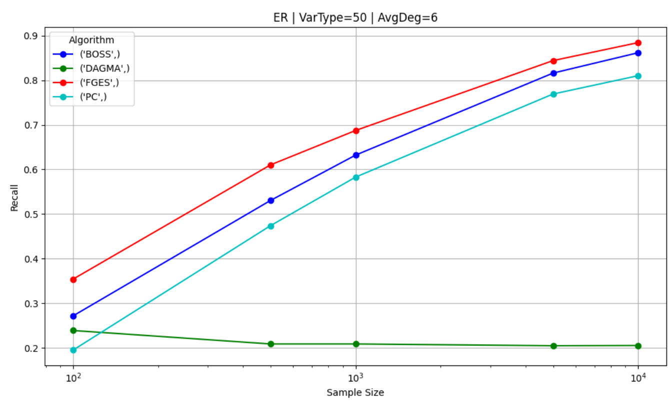

Figure 4. Erdos Renyi Graph, 50 Variables, Average Degree 6. Recall plot

Figure 2, 3, and 4 displays the Precision, Recall, and F1 scores for the random graph baseline.

- **F1 Score:** BOSS, FGES, and PC all demonstrate asymptotic convergence as sample size increases. At $n = 10^4$, BOSS achieves the highest F1 score (≈ 0.92), slightly outperforming FGES (≈ 0.90) and PC (≈ 0.89). DAGMA remains flat with an F1 score around 0.33 across all sample sizes.
- **Precision:** BOSS and PC exhibit high precision, converging to nearly 1.0 at $n = 10^4$. FGES reaches a lower precision plateau of approximately 0.91.
- **Recall:** FGES leads in recall for most sample sizes, reaching $\approx$ 0.88 at $n = 10^4$. BOSS follows closely with a recall of ≈ 0.86, while PC lags slightly behind at ≈ 0.81.

4.2 Performance on Scale-Free Topologies

The following sections detail performance on networks with Scale Free or preferential attachment (hub and sink nodes).

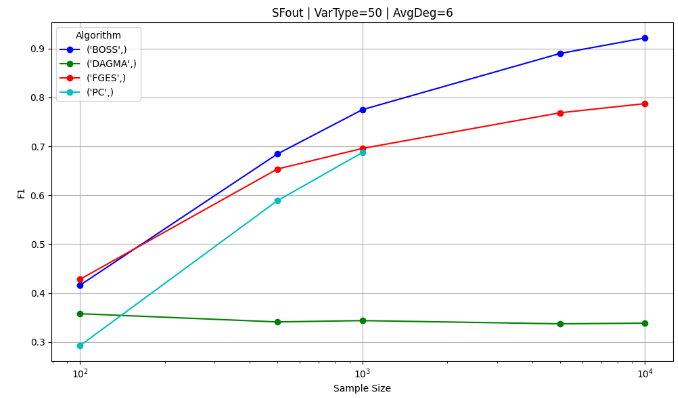

Figure 5. Scale Free Out Graph, 50 Variables, Average Degree 6. F1 plot

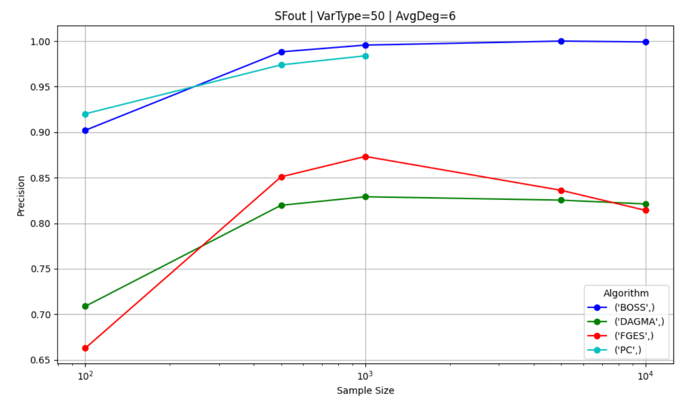

Figure 6. Scale Free Out Graph, 50 Variables, Average Degree 6. Precision plot

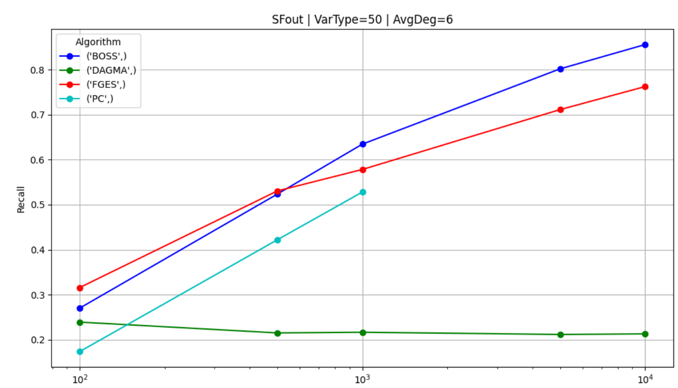

Figure 7. Scale Free Out Graph, 50 Variables, Average Degree 6. Recall plot

4.2.1 Scale-Free Out-Degree (SF-Out) In graphs dominated by hub nodes (Figures 5, 6, and 7):

- **F1 Score:** BOSS shows a marked advantage, reaching an F1 score of ≈ 0.92 at $n = 10^4$. FGES plateaus at ≈ 0.79, and PC reaches ≈ 0.69 (at $n = 10^3$, data for 10^4 not plotted for PC, because, the execution time was capped at 96 hours).

- **Precision:** Similar to the ER case, BOSS and PC maintain near-perfect precision (≈ 1.0) at larger sample sizes. FGES shows a decrease in precision, dropping to ≈ 0.81 at $n = 10^4$.
- **Recall:** BOSS achieves the highest recall (≈ 0.86 at $n = 10^4$). FGES follows with ≈ 0.76. PC shows significantly lower recall, reaching only ≈ 0.53 at $n = 10^3$.

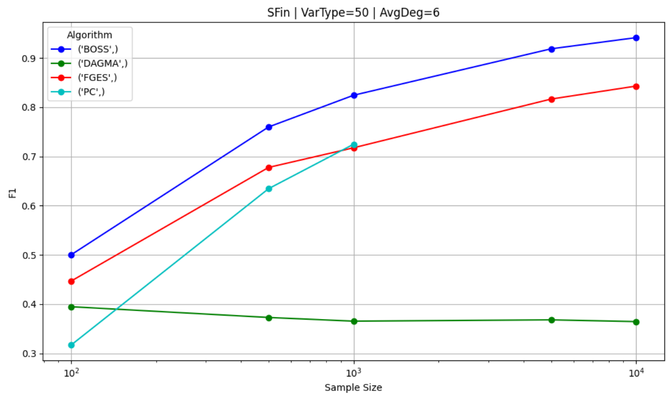

Figure 8. Scale Free Out Graph, 50 Variables, Average Degree 6. F1 plot

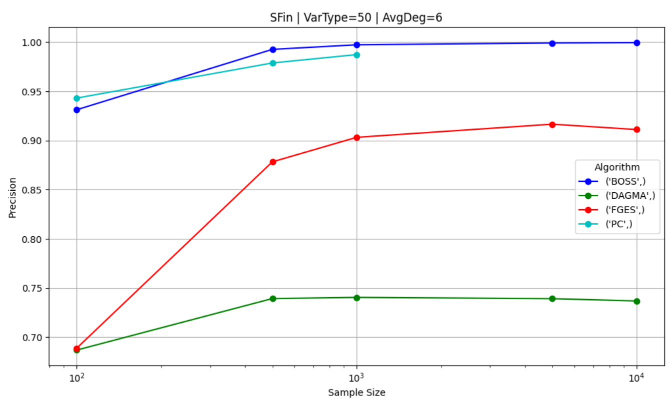

Figure 9. Scale Free Out Graph, 50 Variables, Average Degree 6. Precision plot

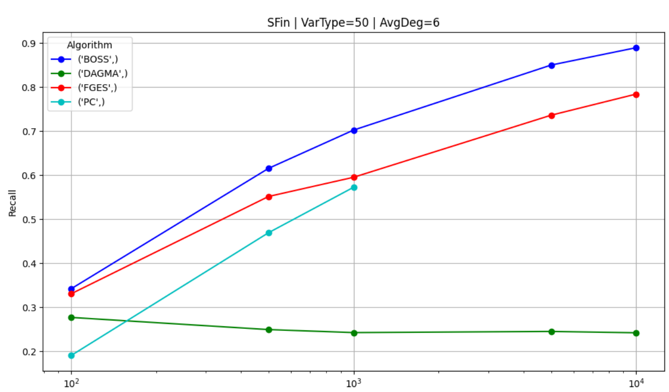

Figure 10. Scale Free Out Graph, 50 Variables, Average Degree 6. Recall plot

4.2.2 Scale-Free In-Degree (SF-In)

In graphs dominated by sink nodes (Figures 8, 9, and 10):

- **F1 Score:** BOSS again outperforms other methods with a final F1 score of ≈ 0.94. FGES reaches ≈ 0.84. PC shows improved performance compared to SF-Out, reaching ≈ 0.72 at $n = 10^3$.
- **Precision:** BOSS and PC converge to precision values of ≈ 1.0. FGES improves slightly compared to the SF-Out case, ending with a precision of ≈ 0.91.
- **Recall:** BOSS leads with a recall of ≈ 0.89 at $n = 10^4$, followed by FGES at ≈ 0.78.

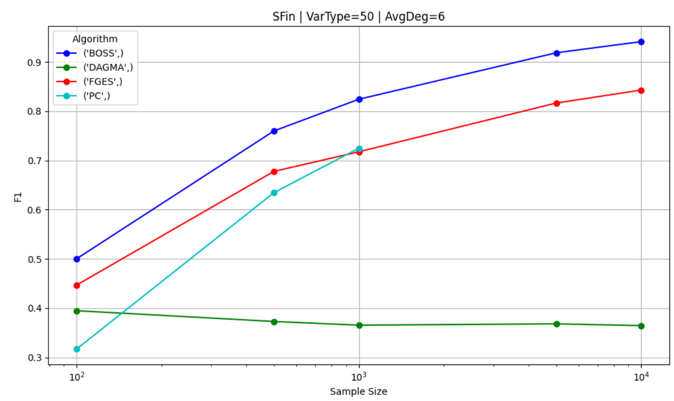

Figure 11. Scale Free Both Graph, 50 Variables, Average Degree 6. F1 plot

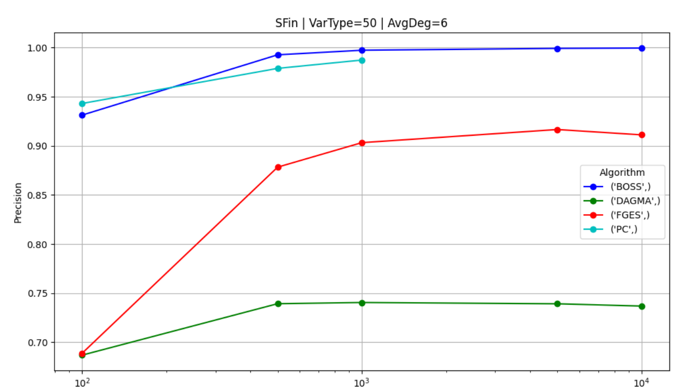

Figure 12. Scale Free Both Graph, 50 Variables, Average Degree 6. Precision plot

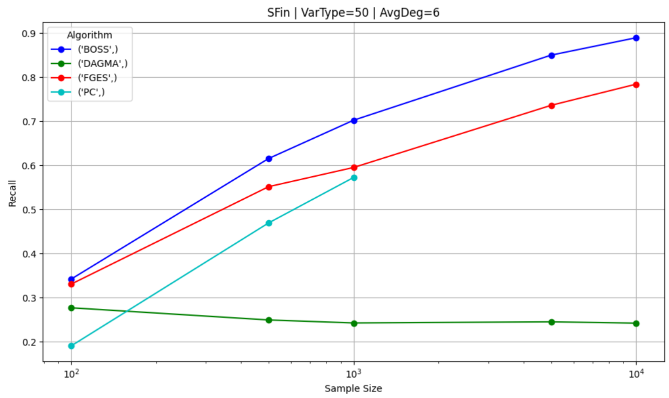

Figure 13. Scale Free Both Graph, 50 Variables, Average Degree 6. Recall plot

4.2.3 Scale-Free Both (SF-Both) In the most complex topology containing both hubs and sinks (Figures 11, 12, and 13):

- **F1 Score:** The divergence between methods is most pronounced here. BOSS achieves an F1 score of ≈ 0.93 at $n = 10^4$. FGES plateaus at ≈ 0.82.
- **Precision:** BOSS maintains a precision of nearly 1.0. FGES precision remains lower, fluctuating between 0.88 and 0.90.
- **Recall:** BOSS demonstrates superior recall (≈ 0.88), effectively recovering edges in both hub and sink structures. FGES reaches a recall of ≈ 0.78.

Across all topologies, DAGMA consistently yields low F1 scores (< 0.40), driven by low precision and recall ($\approx 0.20 - 0.25$) that do not improve with sample size.

5 Discussion

The results of this simulation study highlight a critical dependency between causal discovery performance and the topological structure of the data-generating mechanism. While classical benchmarks on Erdős-Rényi graphs suggest that methods like PC and FGES are highly effective, our findings indicate that these results do not uniformly generalize to scale-free networks, which are more representative of biological and academic citation graphs.

5.1 The robustness of Permutation-based Search

The most consistent finding across all topologies is the superior performance of BOSS. Unlike FGES and PC, which search directly in the space of DAGs or equivalence classes (CPDAGs), BOSS searches over variable orderings. This approach appears particularly advantageous in scale-free networks. In such topologies, identifying the correct causal order of a "hub" node relative to its many neighbors is computationally challenging for local greedy searches. By scoring permutations, BOSS effectively integrates global structural constraints, allowing it to maintain near-perfect precision (≈ 1.0) without the significant recall penalty observed in constraint-based methods. This suggests that for domains known to exhibit power-law distributions (e.g., gene regulatory networks), permutation-based approaches offer the best trade-off between reliability and completeness.

5.2 The Recall Gap in Constraint-Based Methods

The PC algorithm exhibited a distinct vulnerability in Hub-heavy (SF-Out) networks, characterized by a sharp decline in recall. This aligns with the theoretical limitations of the Fisher Z-test in high-density neighborhoods. In a scale-free network, testing conditional independence for a hub node requires conditioning on large sets of variables. As the conditioning set size increases, the statistical power of the test decreases for a fixed sample size, leading to an increased rate of Type II errors (missing true edges). Consequently, PC becomes overly conservative in the presence of hubs, explaining the observed "Precision-Recall gap" where edges found are correct, but many causal links remain undiscovered.

5.3 Precision issues in Greedy Search

While FGES maintained higher recall than PC, it suffered from lower precision, particularly in SF-Out and SF-Both graphs. The greedy nature of the algorithm, which adds edges to maximize a BIC score, appears prone to overfitting the dense neighborhoods surrounding hub nodes. In the absence of strict sparsity penalties, FGES attempts to fit noise as causal influence, resulting in a higher rate of false positives. This makes FGES a riskier choice for exploratory studies in neuroscience, where false positives could lead to incorrect biological hypotheses.

5.4 Challenges with Continuous Optimization

The failure of DAGMA to recover meaningful structures in this discrete setting ($F1 < 0.4$) warrants further investigation. While gradient-based methods have shown promise in continuous domains, their application to discrete causal discovery often requires careful hyperparameter tuning regarding the acyclicity penalty and thresholding. The flatline performance suggests that the non-convex objective function may have trapped the optimization in poor local minima, or that the default initialization is ill-suited for sparse, scale-free graphs.

6 Conclusion and Future Work

6.1 Conclusion

This study aimed to identify the optimal causal discovery algorithm for networks resembling the structural complexity of brain connectivity and citation graphs. By benchmarking four distinct algorithms against Scale-Free and Erdős-Rényi topologies, we demonstrated that algorithm performance is not invariant to network structure.

Our findings establish that **BOSS** is the most robust method for scale-free causal discovery, delivering state-of-the-art F1 scores driven by high precision and superior recall. In contrast, legacy algorithms faced clear trade-offs: **PC** struggled with edge detection (recall) around hub nodes, while **FGES** struggled with edge validity (precision). Consequently, for researchers modeling complex systems where hub-and-sink dynamics are expected, we strongly recommend permutation-based search strategies over traditional constraint or greedy-score approaches.

6.2 Future Work

Future research should aim to expand the scope of these findings in three key directions:

1. **Hyperparameter Sensitivity in Gradient Methods:** Given the theoretical potential of DAGMA, a more rigorous sweep of its sparsity regularization (λ) and thresholding parameters is necessary to determine if its poor performance is intrinsic to the method or a result of suboptimal tuning.

2. **High-Dimensional Scalability:** Our study focused on graphs with $|V| = 50$. As brain networks often involve hundreds or thousands of regions of interest (ROIs), future work must assess the computational scalability of BOSS, as permutation searches can become prohibitively expensive in high-dimensional spaces.
3. **Real-World Validation:** Finally, we intend to apply these insights to empirical datasets, specifically fMRI time-series data and open-source citation networks. Validating whether the edges discovered by BOSS in these domains correspond to known biological pathways or academic influence flows will be the ultimate test of its practical utility.

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